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Based on [2501.09498] with Ma-Ke Yuan and Yang Zhou

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Motivation

Weyl anomaly governs the **entanglement structure** and **ground state energy** of CFTs

2d CFT $E = -\frac{c}{12}$ $S_n = \frac{c}{6}\left(\frac{1}{n}+1\right)\log\frac{l}{\epsilon}$

- ➤ Still valid in the presence of defects?
- ➤ Non-perturbative way to study defects?

Key Results

Closed formulae for surface defect contribution to twisted Rényi entropy and Casimir energy in 6d A_{N-1} (2,0) SCFTs, **controlled by defect Weyl anomaly**

$$S_n = \frac{2b-d_2}{6}\left[\frac{r_1 r_2}{2}\left(\frac{1}{n}-1\right)+1\right]\log\frac{l}{\epsilon}$$

$$E_{\mathcal{R}} = -\frac{1}{\omega_1}\left[\frac{d_2-b}{6}\omega_2\omega_3 + \frac{2b-d_2}{24}\sigma_1\sigma_2\right]$$

Non-perturbative method: holography, localization & anomaly polynomial

Future Direction

- Defects of other co-dim. & in different bulk theories
- High temperature universality (Cardy formula) in the presence of defects